

Question	Answer	Marks
6(a)	series charges: $Q_s = Q_1 = Q_2$	B1
	series p.d.s: $V_s = V_1 + V_2$	B1
	parallel charges: $Q_s = Q_1 + Q_2$	B1
	parallel p.d.s: $V_s = V_1 = V_2$	B1
6(b)(i)	$E = \frac{1}{2} CV^2$	C1
	p.d. = $[(2 \times 19 \times 10^{-3}) / (470 \times 10^{-6})]^{\frac{1}{2}}$ $= 9.0 \text{ V}$	A1
6(b)(ii)	$E = Q^2 / 2C$ or $C = Q / V$	C1
	$Q = (19 \times 10^{-3} \times 2 \times 470 \times 10^{-6})^{\frac{1}{2}}$ or $Q = 470 \times 10^{-6} \times 9.0$ $Q = 4.2 \times 10^{-3} \text{ C}$	A1
6(b)(iii)	total charge unchanged	C1
	total capacitance = $(470 + 180) \times 10^{-6} \text{ (F)}$	C1
	$E = Q^2 / 2C = (4.23 \times 10^{-3})^2 / (2 \times 650 \times 10^{-6})$ (= 0.014 J) $E = 14 \text{ mJ}$	A1

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